IOP Conference Series:
Earth and
Environmental Science

PAPER • OPEN ACCESS

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To cite this article: Nursyafawati Bakhari *et al* 2025 *IOP Conf. Ser.: Earth Environ. Sci.* **1548** 012017

Insulation performance of the coconut husk advancing aerogel for building material

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Abstract. The urgency for energy-efficient in buildings, urges the search for the alternative sustainable insulation material to substitute the. conventional environmentally detrimental options made from synthetic polymers, fiberglass or rock wool. Biomass resources with their inherent insulating properties present a promising alternative in recent studies. To enhance the biomass's insulation performance, advancing the biomass into aerogel form has become attractive owing to its low density and thermal conductivity. In this study, the fabrication of coconut husk aerogel for its thermal insulation capacity is investigated under the varying concentration of the coconut husk mix with constant concentration of sodium alginate and phytic acid. The morphological and chemical composition of the coconut husk aerogel were analysed using scanning electron microscopy with energy-dispersive X-ray spectroscopy (SEM-EDX), Brunauer–Emmett–Teller (BET) and Fourier-transform infrared (FTIR) spectroscopy. The result depicted the increment of the 20% for the porous structure while the FTIR and EDX data exhibits the decreasing of the carbon and oxygen content averagely 25% confirming the cross-linking from the additives. Measurement of the thermal conductivity were performed to quantitatively assess the insulation properties, with an average thermal conductivity of approximately 0.15 W/m.K. This value is markedly lower than the wood material (0.4 W/m.K) highlighting the superior insulation of the coconut husk aerogel. Conversely, 10% of the coconut husk in coconut husk aerogel production was attributed with the increased density and decreased porosity of the material ultimately diminishing its insulating properties. This finding further supports the effectiveness of the other formulations.

1. Introduction

The past decade has witnessed a pronounced surge in global energy consumption, a trend propelled by rapid urbanization and demographic expansion. Within this context, the building sector constitutes a major energy end-use sector, accounting for approximately 30% of global final energy consumption and 26% of global energy-related emissions (International Energy Agency, 2023). Mitigating this footprint necessitates a strategic focus on enhancing building energy efficiency, where high-performance thermal insulation plays a critical role. By effectively retarding heat transfer, these materials significantly reduce the operational energy demand for mechanical heating and cooling.

Consequently, the adoption of environmentally benign insulation alternatives has become imperative, offering a dual benefit: curtailing energy consumption and alleviating associated environmental burdens (Shahee et al., 2024). The current market offers a spectrum of insulation materials, which are generally classified into three categories: natural, synthetic, and recycled options (Rojas et al., 2019). Despite the prevalent use and proven thermal efficacy of synthetic options such as fiberglass and expanded polystyrene, their dominant market position is being progressively re-evaluated amid growing environmental apprehensions. The main concerns centralized on their lifecycle environmental impact, which encompasses energy-intensive manufacturing and complications surrounding end-of-life disposal stemming from their non-biodegradable and often non-recyclable nature (Abu-Jdayil et al., 2019).

Coconut husk represents a compelling agricultural waste product that is increasingly recognized for its potential as a sustainable feedstock for eco-friendly insulation materials. Its appeal lies in both its inherent thermal stability and its renewable nature (Stelte et al., 2022). The viability of this biomass is further reinforced by its regional abundance; in nations such as Malaysia under the hot-humid climate's growth capture a sustained increasing of the coconut production output between 2014 and 2019 by 10% increment while its waste materials are still underutilized (Zakaria et al., 2022). Research studies highlighted that the raw coconut husk fibers without any pretreatment exhibit notable insulating properties, capable of moderating interior climates and diminishing the energy required for mechanical heating and cooling (Iwaro and Mwasha, 2019).

To enhance these natural properties, the fibers can be incorporated into an aerogel matrix. Within this matrix, sodium alginate, a natural biopolymer derived from brown seaweed acts as structural binder. Cross-linked this biopolymer with divalent cations via a sol-gel process, yields a robust, porous hydrogel. Subsequent freeze-drying of this structure produced a solid aerogel with low thermal conductivity. The resulting fabrication effectively minimizes heat transfer, thereby demonstrated an effective thermal insulator with considerable promise in reducing energy consumption in the buildings (Wang and Lu, 2023; Giuma et al., 2024). Since these substances derived from renewable sources and decomposes naturally, the resulting material represents a sustainable substitute for traditional insulation products in market (Deng et al., 2023).

This study purposely examines the thermal properties of the fabricated coconut husk aerogel and assess its viability as insulating materials. By combining the naturally insulative characteristics of coconut husk with aerogel fabrication techniques, the research aims to create a renewable and high-performing alternative to conventional options. The use of coconut husk not only valorizes agricultural waste but also aligns with sustainable building practices. However, challenges such as low mechanical durability, sensitivity to moisture, and production scalability must be addressed. To tackle these limitations, the study incorporates material enhancement and performs thermal characterization to assess their potential for implementation in energy-saving building applications.

2. Materials and methods

2.1 Experimental materials and chemicals

The raw coconut husk was collected from a local marketplace in Ketereh, a district situated in the eastern region of Kelantan, Malaysia. To minimize the moisture content, the collected coconut husk was subjected to oven drying at 60°C for one hour and subsequently milled into fine powder,

IOP Conf. Series: Earth and Environmental Science **1548** (2025) 012017 doi:10.1088/1755-1315/1548/1/012017

producing approximately 125 grams of processed material. This powder was sealed in a ziplock bag and stored at room temperature without further chemical pretreatment for later experimental use. The primary chemical reagent including phytic acid and sodium alginate were procured from Tiongnam Logistic Solution Sdn Bhd. Distilled water served as the solvent in preparing the samples to preserve reagent purity and maintan procedural consistency.

2.2 Experimental fabrication of the coconut husk aerogel

The coconut husk aerogel was fabricated via a freeze-drying technique follow on an establish protocol by Ho et al. (2021). In the present work, the coconut husk volume served as the primary variable, with tested concentration ranging from 5% to 25% (w/v). Meanwhile, phytic acid and sodium alginate were consistently added at a fixed concentration of 5% (w/v) and 20% (w/v), respectively, for every formulation. The mixture was homogenized with a magnetic stirrer for three minutes, transferred to a 100 mL beaker. The sample were then placed in a sealed desiccator for over five days to facilitate the cross-linking and gelation while preventing solvent evaporation. the resultant gels were then subjected to freeze-drying at -55°C for five additional days to sublime the aqueous phase, yielding a dry, porous aerogel structure with thermal insulating features.

2.3 Characterisation of the coconut husk aerogel

The surface morphology and compositional properties of the coconut husk aerogel were investigated employed the scanning electron microscopy integrated with energy-dispersive X-ray spectroscopy (SEM-EDX). Analyses were performed at an accelerating voltage of 10 kV and a working distance of 9.7 mm. Elemental identification was achieved using a standardless ZAF correction method, with signal acquisition facilitated by a secondary electron detector (SED). The textural properties, including porosity and specific surface area, were determined from nitrogen adsorption-desorption isotherms measured at 77 K. The Brunauer–Emmett–Teller (BET) method was applied to measure the specific surface area, while the t-plot method was used to derive the micropore volume and surface area. Meanwhile, the chemical functional groups within the coconut husk aerogel were identified using Fourier-transform infrared (FTIR) spectroscopy. The spectra were acquired in the transmittance mode from 400 to 4000 cm^{-1} employed potassium bromide (KBr) pellets to ensure optimal signal clarity. In addition to this morphological and chemical characterization, the thermal properties of the aerogel were then evaluated to determine its stability and efficacy as a thermal insulator for building applications.

2.4 Thermal conductivity measurement

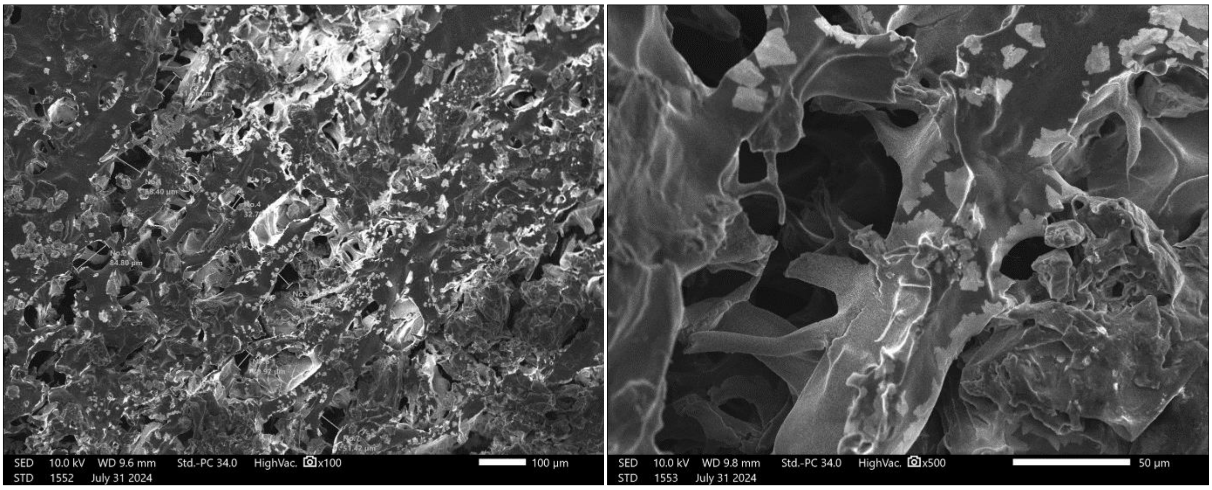
The thermal conductivity measurement of the coconut husk aerogel samples was determined using a Hot Disc TPS 2500S thermal constant analyzer. This specific instrument was selected due to its ability to accommodate samples measuring 20 mm in thickness, aligned with a standard dimension for evaluating building insulation materials (Huang et al., 2020). Owing to the sensor diameter of 13 mm smaller than the sample thickness, the measurement configuration required sandwiching the sensor between two aerogel specimens to guarantee complete coverage. Subsequently, the transient plane source (TPS) customized via the instruments's proprietary software, was adopted to assess the thermal conductivity and specific heat capacity of each fabricated samples. In order to affirm the realibilty of the acquired data, all measurements were performed in triplicate.The mean values alongside the standard deviation were calculated for each data obtained.

3. Results and discussion

3.1 Characterisation of coconut husk aerogel

3.1.1 Surface morphology

SEM-EDX analysis was employed in evaluating the aerogel's microporosity composed of coconut husk. This combined analytical approach offering essential insight to assess the material's potential as an efficient thermal insulator. It has been discovered that a non-uniform but connected porous structure is depicted in SEM images at 100x and 500x resolution of figures 1 (a) and (b), demonstrating good thermal insulation features. The role of sodium alginate, which promotes cross-linking inside the coconut husk matrix during gel formation and creates a more stable and porous network, influences this structure (Cao et al., 2021). The EDX data in table 1 further confirm the elemental uniformity and successful integration of the components, with the table presenting the composition of coconut husk and coconut husk aerogel.



(a) (b)

Figure 1. SEM images of coconut husk aerogel; (a) 100x magnification and (b) 500x magnification.

Table 1. Elemental composition of raw coconut husk and coconut husk aerogel.

Sample	Carbon (C)		Oxygen (O)	
	Mass (%)	Atoms	Mass (%)	Atoms
Coconut husk	54.54	61.51	45.46	38.49
Coconut husk aerogel	34.13	46.31	38.10	38.81

The results reveal a high carbon content (54.54%) and a substantial oxygen presence (45.46%), indicating a strong carbonaceous framework suitable for chemical functionalization. Studies suggest that carbon content ranging from 50% to 71% is optimal for fabricating porous materials with favorable physicochemical properties (Xu et al., 2020). Moreover, Wu at al. (2022) suggested that carbon-based materials are known for low thermal conductivity, while oxygen-containing functional groups improve phonon scattering, both of which contribute to superior insulation performance Post-conversion to aerogel, the carbon and oxygen levels decline to

34.13% and 38.10%, respectively likely due to processing variables such as drying technique, gelation conditions, or precursor inconsistencies that contribute to the oxidation of carbon in the air (Zheng et al., 2024).

3.1.2 Textural parameters

The textural parameters of the coconut husk aerogel, emphasizing both on pore structure and surface area, are presented in table 2. The value of the specific surface area of the coconut husk aerogel at 5.1283 m^2/g, representing a significant reduction form the 11.45 m^2/g observed in the untreated coconut husk. The aerogel's result denoted the value of well below the typical range of 200 – 700 m^2/g reported for aerogels (Azadi and Dinari, 2023). This reduction may be due to variations in processing methods such as gelling agent composition, drying conditions, or insufficient activation of precursor materials. The diminished surface area suggests a denser or more compact structure in the aerogel compared to the raw coconut husk, possibly caused by structural reorganization during synthesis. This is supported by Poornachandhra et al. (2023) stated that extensive process exhibits low measured surface area even materials become more porous at larger scale.

The result denoted that the gelation process did not create a new microporous network, instead it extended the existing structure of the coconut husk. The gelation processed dissolved some of the lignin and hemicellulose and thus enlarge the natural pores and yielding a more monolithic solid foam (Anuchi et al., 2022). Hence, it can be seen from the decreasing of the surface area and increasing in the porosity. The average pore diameter increased from 2.261 nm of the raw husk to 2.852 nm in the aerogel, indicating the formation of a more open mesoporous network. According to Yang et al. (2023), such mesopores, typically between 2 nm and 50 nm, are beneficial for insulation by trapping air and limiting heat transfer. Additionally, both samples displayed similar bulk densities of approximately 1.000 g/cm^3.

Despite the reduced surface area, SEM analysis confirmed an interconnected porous architecture, reinforcing the aerogel's thermal insulation potential. The increase in solid conduction due to the lower surface area is a trade-off that was likely inherent to the bio-based synthesis process. However, the final material remains an effective insulator. The synergy of mesoporosity and structural coherence supports the viability of coconut husk aerogels as sustainable insulating materials.

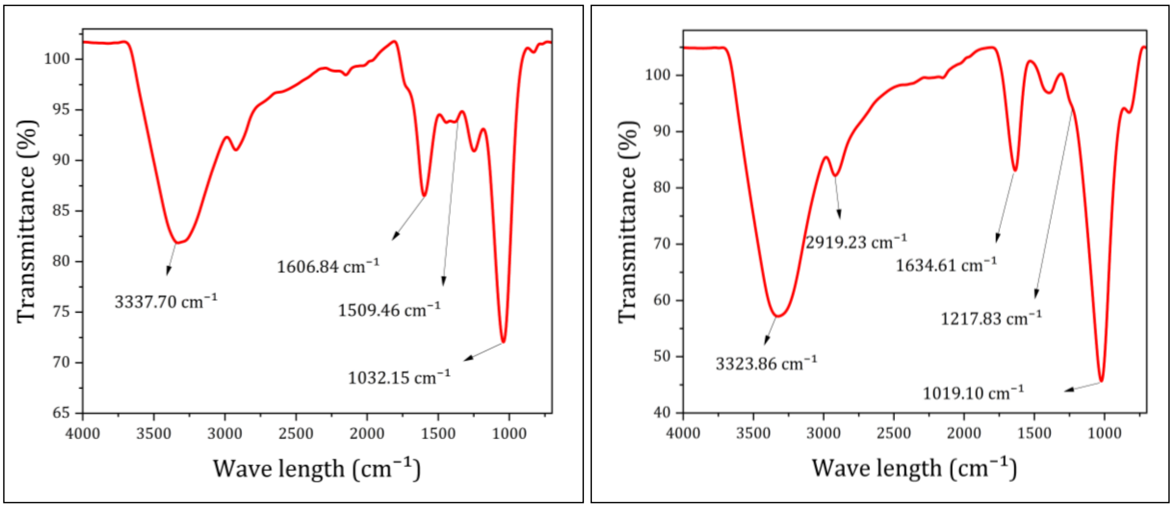
Table 2. Textural analysis of coconut husk and coconut husk aerogel.

Sample	Parameter		
	BET surface area (m^2/g)	Pore size (nm)	Bulk density (g/cm^3)
Coconut husk	11.45	2.261	1.000
Coconut husk aerogel	5.130	2.852	1.000

3.1.3 Identification of the functional group

FTIR analysis of coconut husk and coconut husk aerogel in figures 2 (a) and (b) reveal notable chemical differences across a wavelength range of 4000 – 500 cm^{-1}. The coconut husk spectrum displays prominent peaks at 3337.70 cm^{-1} (O–H), 1606.84 and 1509.46 cm^{-1} (C=O and C=C), and 1032.15 cm^{-1} (C–H), indicating moisture content and lignocellulosic components, including cellulose, hemicellulose, and lignin (Kim et al., 2021). In contrast, coconut husk aerogel shows peaks at 3323.86 cm^{-1} (O–H), 2919.23 cm^{-1} (C–O–C), 1634.61 cm^{-1} (P=O), and 1217.83 cm^{-1}

(C=O), revealing reduced hydroxyl groups and the formation of new bonds due to interactions with phytic acid and sodium alginate (Zhang et al., 2018). The new P=O and C–O–C bonds signify structural modifications that enhance aerogel thermal resistance and moisture interaction, supporting coconut husk aerogel's potential as a stable and efficient insulating material.



(a) (b)

Figure 2. FTIR spectra; (a) coconut husk, and (b) coconut husk aerogel.

3.2 Thermal conductivity of coconut husk aerogel

Thermal conductivity serves as a key metric in evaluating insulation performance for energy-efficient construction materials (Cao et al., 2021). In this study, thermal conductivity was assessed across samples containing varying concentrations of coconut husk ranging from 5% to 25%, with testing conducted at a stable temperature of 25.5 °C. The results, detailed in table 3, reveal that a 5% coconut husk composition achieved a thermal conductivity of 0.1318 ± 0.000535 W/m.K. The highest conductivity was observed at 10% concentration with 0.1937 ± 0.00295 W/m.K, followed by a gradual decline at 15% (0.1771 ± 0.001494 W/m.K), 20% (0.1574 ± 0.00566 W/m.K), and 25% (0.1341 ± 0.000914 W/m.K). This trend suggests that increased coconut husk content beyond 10% may enhance the material's structure, improving its thermal resistance.

Table 3. Various coconut husk concentration on the thermal conductivity.

Coconut husk (%)	Temperature (°C)	Thermal conductivity (W/m.K)
5	25.5	0.1341 ± 0.000535
10	25.5	0.1937 ± 0.00295
15	25.5	0.1771 ± 0.001494
20	25.5	0.1574 ± 0.00566
25	25.5	0.1341 ± 0.000914

Figure 3 illustrates the thermal conductivity values measured at varying coconut husk concentrations (5% to 25%) and compares them with standard benchmarks for common insulation materials. As reported by Mededji et al. (2024), wood (0.15 – 0.40 W/m.K) serves as a representative good insulator and glass wool (0.03 – 0.05 W/m.K) exemplifies an excellent thermal insulator. Overall, the coconut husk aerogel recording averagely at 0.15 W/m.K, almost half the value recorded for wood (0.40 W/m.K) resulted a significant enhancement in thermal performance. Among the samples, the 10% coconut husk composition exhibited the highest thermal conductivity of approximately 0.1937 W/m.K, exceeding the upper limit of the wood benchmark. In contrast, the 5% and 25% compositions recorded lower values at 0.1318 W/m.K and 0.1341 W/m.K, respectively, falling within the typical wood range and suggesting comparatively better insulation characteristics. The samples with 15% and 20% coconut husk demonstrated intermediate thermal conductivities of around 0.1771 W/m.K and 0.1574 W/m.K. None of the formulations approached the thermal performance of glass wool, whose benchmark remains significantly lower.

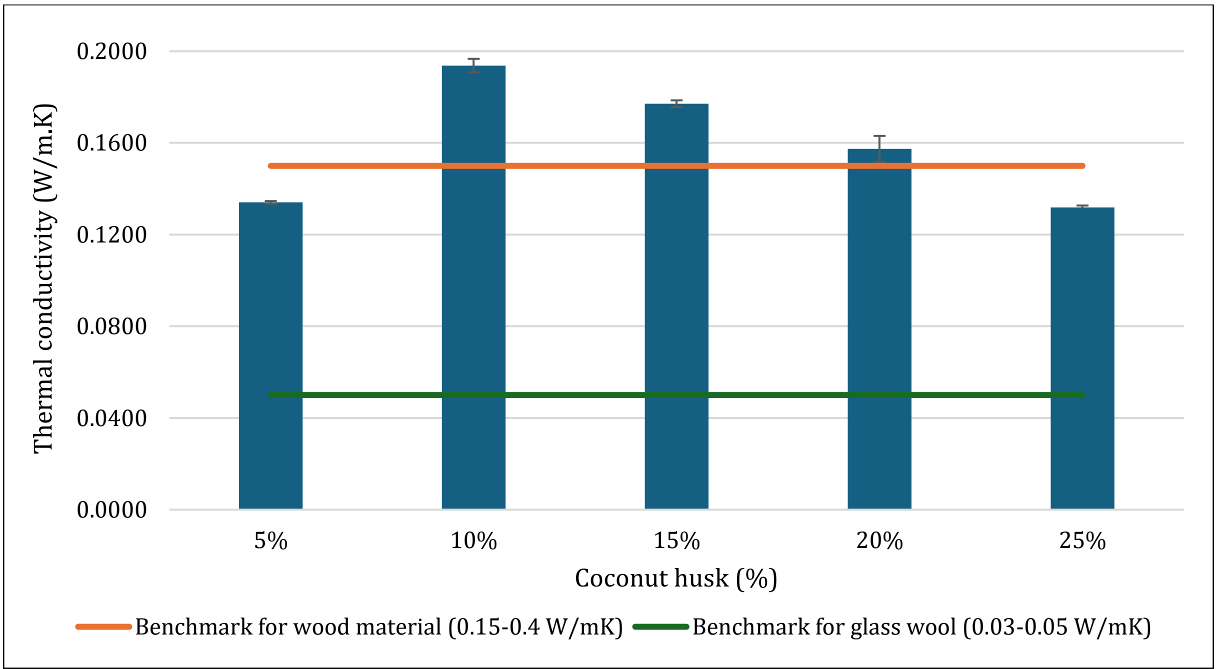


Figure 3. Comparison of thermal conductivity across varying coconut husk concentrations with reference to typical values for wood and glass wool.

These findings highlight that while coconut husk aerogel can meet or approach conventional wood insulation standards, the material still fall short of matching high-performance materials like glass wool. The observed trend suggests that increased husk concentration beyond the optimal point alters the microstructure, possibly increasing thermal bridging pathways and reducing insulating capacity. The increase in insulation efficiency at the 15 – 25% range of intermediary of biomass concentrations is most likely the result of structural alterations that elevate the biomass's porous internal microstructure, thus diminishing the material's heat transfer interval. It has been shown that increased porosity capturing air pockets functioning as thermal barriers consequently reduce thermal conductivity in biobased insulation materials (Li et al., 2024; Wang et al., 2018). On the other hand, the 10% concentration levels of thermal conductivity are more analogous to conventional wood, implying limited insulation efficacy. These findings suggest that the aerogel's insulating properties can be enhanced by integrating with the wood's constituent materials.

4. Conclusion

This study fabricated coconut husk aerogel with enhanced thermal insulation properties by systematically varying the coconut husk volume and maintaining the volume of sodium alginate and phytic acid. A series of different mixtures of the coconut husk in the mixture were studied particularly the thermal performance. Surface morphological and elemental composition investigation conducted via SEM-EDX, BET, and FTIR, revealed coconut husk aerogel exhibit a porous structure and dominantly composed of lignocellulosic functional groups with thermal insulation capabilities falling within a defined performance range. Notably, the coconut husk aerogel fabricated with over a 10% volume of coconut husk in the fabrication process, exhibited remarkable thermal insulation features, highlighting their significant potential for energy-conserving building. In conclusion, coconut husk aerogel possesses a great potential and served as a green alternative for conventional insulation materials. Future work should focus on exploring the various advances fabrication methods which resulted greater surface area gain and productively assessing the material's durability over time under wide range of environmental circumstances.

Acknowledgement

This study was supported by Ministry of Higher Education, Malaysia under the grant FRGS-EC (RDU243720) and Universiti Malaysia Pahang Al-Sultan Abdullah through the internal research grant (RDU220319).

References

Abu-Jdayil, B., Mourad, A-H., Hittini, W., Hassan, M. & Hameedi, S., 2019. Traditional, state-of-the-art and renewable thermal building insulation materials: An overview. Construction and Building Materials, 214, pp.709–735. doi:10.1016/j.conbuildmat.2019.04.102

Anuchi, S., Campbell, R. & Hallett, J.P., 2022. Effective pretreatment of lignin-rich coconut wastes using a low-cost ionic liquid. Scientific Reports, 12, pp.6108. doi:10.1038/s41598-022-09629-4

Azadi, E. & Dinari, M., 2023. Green synthesis, characterization, and properties of carbon aerogels. In: S.U. Islam & C.M. Hussain, eds. ACS Symposium Series. Washington, DC: American Chemical Society, pp.1–23. doi:10.1021/bk-2023-1441.ch001

Cao, M., Liu, B-W., Zhang, L., Peng, Z-C., Zhang, Y-Y., Wang, H., Zhao, H-B. & Wang, Y-Z., 2021. Fully biomass-based aerogels with ultrahigh mechanical modulus, enhanced flame retardancy, and great thermal insulation applications. Composites Part B: Engineering, 225, pp.109309. doi:10.1016/j.compositesb.2021.109309

Deng, T., Li, H., Li, Y., Jiang, C., He, Y., Yang, T., Zhu, L. & Xie, L., 2023. Environment friendly biomass composite aerogel with reinforced mechanical properties for thermal insulation and flame retardancy application. Polymer Engineering & Science, 63(12), pp.4084–4097. doi:10.1002/pen.26509

Giuma, A., Khalil, H.A., Yahya, E.B., Sukeksi, L., Alfatah, T., Nurazzi, N.M., Jaber, M. & Surya, I., 2024. Green thermal insulators: A review into the role of biopolymer-based aerogels in thermal insulation applications. Polymer Engineering & Science, 64(10), pp.4611–4629. doi:10.1002/pen.26888

Ho, K.H., Nguyen, P.X.T., Do, N.H.N., Le, K.A. & Le, P.K., 2021. Green fabrication of bio-based aerogels from coconut coir for heat insulation and oil removal. Chemical Engineering Transactions, 88, pp.709–714. doi:10.3303/CET2188118.

Huang, H., Zhou, Y., Huang, R., Wu, H., Sun, Y., Huang, G. & Xu, T., 2020. Optimum insulation thicknesses and energy conservation of building thermal insulation materials in Chinese zone of humid subtropical climate. Sustainable Cities and Society, 52, 101840. doi:10.1016/j.scs.2019.101840

International Energy Agency (IEA)., 2023. Energy sector CO_2 emissions include emissions from energy combustion and industrial processes; Buildings – topics. IEA. Retrieved August 31, 2025, https://www.iea.org/energy-system/buildings#tracking

Iwaro, J. & Mwasha, A., 2019. Effects of using coconut fiber–insulated masonry walls to achieve energy efficiency and thermal comfort in residential dwellings. Journal of Architectural Engineering, 25(1), 04019001. doi:10.1061/(ASCE)AE.1943-5568.0000341

Li, M., Wang, F., Ouyang, S., Liu, Y., Hu, Z., Wu, Y., Qian, J., Li, Z., Wang, L. & Ma, S., 2024. A comprehensive review on preparation and functional application of the wood aerogel with natural cellulose framework. International Journal of Biological Macromolecules, 133340. doi:10.1016/j.ijbiomac.2024.133340

Mededji, D.S., Sogbochi, E., Djossou, A.A., Cherkaoui, O., Fagbemi, L.A. & Sohounhloue, D.C.K., 2024. Characteristics of insulating panels realised from coconut palm fibres for low-temperature thermal insulation. International Journal of Environment and Climate Change, 14(5), pp.324–335. doi: 10.9734/ijecc/2024/v14i54192

Poornachandhra, C., Jayabalakrishnan, R.M., Prasanthrajan, M., Balasubramanian, G., Lakshmanan, A., Selvakumar, S. and John, J.E., 2023. Cellulose-based hydrogel for adsorptive removal of cationic dyes from aqueous solution: isotherms and kinetics. RSC advances, 13(7), pp.4757-4774

Rojas, C., Cea, M., Iriarte, A., Valdés, G., Navia, R. & Cárdenas, J.P.R., 2019. Thermal insulation materials based on agricultural residual wheat straw and corn husk biomass, for application in sustainable buildings. Sustainable Materials and Technologies, 20, e00102. doi:10.1016/j.susmat.2019.e00102

Shahee, A., Abdoos, M., Aslani, A. & Zahedi, R., 2024. Reducing the energy consumption of buildings by implementing insulation scenarios and using renewable energies. Energy Informatics, 7(1), 18. doi:10.1186/s42162-024-00311-9

Stelte, W., Reddy, N., Barsberg, S. & Sanadi, A.R., 2022. Coir from coconut processing waste as a raw material for applications beyond traditional uses. BioResources, 18(1). doi:10.15376/biores.18.1.Stelte

Wang, Y. & Lu, Y., 2023. Sodium alginate-based functional materials toward sustainable applications: Water treatment and energy storage. Industrial & Engineering Chemistry Research, 62(29), pp.11279–11304. doi: 10.1021/acs.iecr.3c01082

Wang, X., Liu, H., Wang, L., Liu, B. & Wang, L., 2018. Optimization of thermal insulation performance for the porous materials. Heat Transfer Engineering, 39(17–18), pp.1573–1581. doi: 10.1080/01457632.2017.1369847

Wu, K., Zhou, Q., Cao, J., Qian, Z., Niu, B. & Long, D., 2022. Ultrahigh-strength carbon aerogels for high temperature thermal insulation. Journal of Colloid and Interface Science, 609, pp.667–675. doi:10.1016/j.jcis.2021.11.067

Xu, Y., Liu, Y., Chen, S. & Ni, Y., 2020. Current overview of carbon fiber: Toward green sustainable raw materials. BioResources, 15(3), pp.7234–7259. doi:10.15376/biores.15.3.Xu

Yang, J., Wu, H., Liang, Y., Cen, J. and Zhang, X., 2023. A comparative assessment of different aerogel-insulated building walls for enhanced thermal insulation performance. Gels, 9(12), p.943

Zakaria, M.H., Mat Amin, M.Z., Ahmad, M.F. & Ahmad Dani, M.S., 2022. Market potential and competitiveness assessment of Malaysian coconut-based products. Econ. Technol. Manag. Rev., 18, pp.11–22

Zhang, L., Tu, L.Y., Liang, Y., Chen, Q., Li, Z.S., Li, C.H. & Li, W., 2018. Coconut-based activated carbon fibers for efficient adsorption of various organic dyes. RSC Advances, 8(74), pp.42280–42291. doi: 10.1039/c8ra08990f

Zheng, C., Li, X., Yu, J., Huang, Z., Li, M., Hu, X. & Li, Y., 2024. Biomass-derived lightweight SiC aerogels for superior thermal insulation. Nanoscale, 16(9), pp.4600–4608. doi:10.1039/D3NR06076D